PH.45320 **FALL 2026**

Static Cosmology

Lecture 3: Static Radiative Transfer

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Overview

The transfer equation includes absorption, emission, and angular redistribution in an infinite static domain.

1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$dI_\nu/ds = -\alpha_\nu I_\nu + j_\nu + \int d\Omega' R(\Omega, \Omega') I_\nu(\Omega')$$

2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$S_\nu = j_\nu/\alpha_\nu$$

3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$I_\nu(\infty) < \infty for \alpha_\nu > 0$$

4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Jeong, ch. 3
- Radiative Transfer Tables RT-9
- ETHERMAP ray guide